Artificial Intelligence for robotics

Part 2: the neuron

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Artificial Intelligence for Robotics: the neuron

- **Artificial intelligence trie to implement several properties of human or animal intelligence**
- **The only known form of intelligence emerge from the most incredible, but also the most mysterious organ of living beings: the brain**

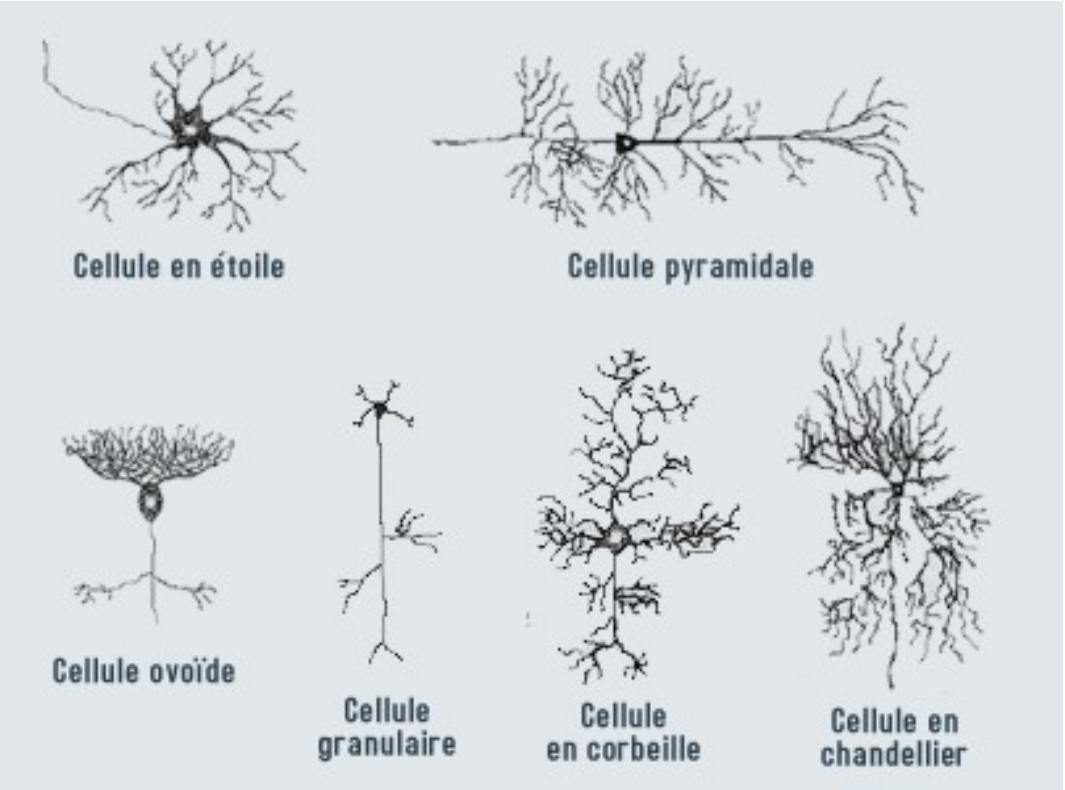


  - **Case of human brain:**
    - ~80-100 billion neurons
      - *As much glial cells (many of which are astrocyts)*
    - ~ 1,5 x 10^{14} connections
    - 10.000 to 50.000 neurons and 100 to 500 millions connections per mm^3
    - Many specialized areas
  - Even animals cannot be fully mapped and simulated (especially due to the complexity of neurons and their connections)

- **Let's analyze the fundamental component of the brain : the neuron**

Artificial Intelligence for Robotics: the neuron

Main neuron types (>200 types) :



New types are regularly discovered:

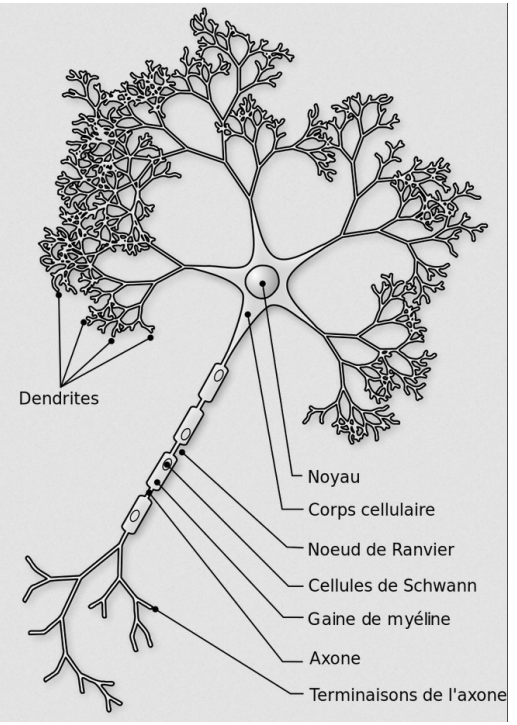
→ Rosehip in 2018

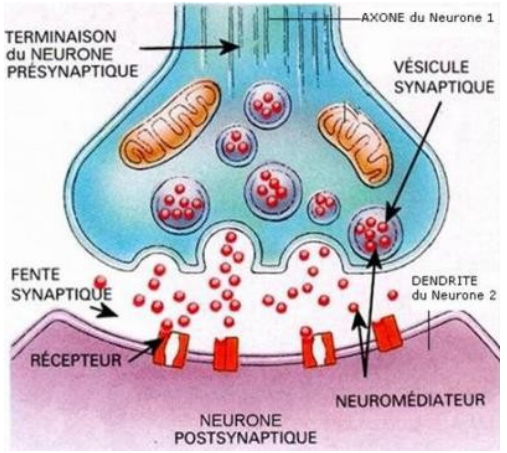
→ Campana in 2022

Artificial Intelligence for Robotics: the neuron

- Anatomy of a unipolar neuron:
  - A body (also called soma)
  - Inputs: the *dendrites* (around 7000 per neuron)
  - Output: a unique *axon*

- Neurons are connected through *synapses*
  - A synapse links the axon of a neuron and the dentrite of another neuron
  - Signal transmission is unidirectional
    - But sometime bi-directional in some types
  - No direct contact between neurons
    - Emitter side:
      - the neuron produces neurotransmitters
    - Receiver side:
      - the neuron detects the neurotransmitters
  - The more receivers there are, the stronger the signal received
  - Each synapse can contains multiple types of neurotransmitters.

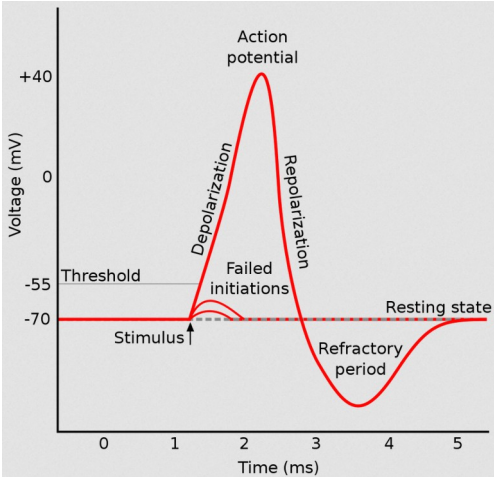




Artificial Intelligence for Robotics: the neuron

- A neuron generates pulses of 1 to 2 milliseconds
  - Its activity can be defined as the frequency of pulse emission.

- A neuron has a potential of -70mV relative to ambient environment (rest tension) due to the presence of ions (Na^+, K^+, Cl^- et Ca^{2+})
- When receiving a synaptic signal, its tension increases or decreases (depending of neurotransmitter type)
- When the tension becomes greater than the treshold of -55 mV :
  - the membrane channels open to allow ions to pass through (depolarization)
  - A signal (action potential) of +100 mV is transmitted through the axon
  - The synaptic vesicles release neurotransmitters
  - The neuron returns to its resting potential (-70 mV)



- The signal (and thus information) is thus related to the synaptic connection strength between neurons of the network

Artificial Intelligence for Robotics: the neuron

Neuron learning mechanism

- Most common mechanism among neurons: principle of Hebb (Donald Hebb , 1949)
  - Neurons firing simultaneously reinforce their connection.
  - → When two connected neurons depolarize simultaneously, the synapse increases the number of neurotransmitter vesicules.
    - Consequence: when the emitter neuron's signal has a great incidence on the activation of receiver neuron, the connection must be reinforced.
- Another mechanism: the long-term polarization, mostly used by neuron of the hippocamp (area responsible of long term memory and navigation)
  - In case oh high frequency stimulation, the synaptic connection is reinforced for a long period of time